

System Monitoring and Performance Tuning Continue...

When monitoring memory usage in Linux, several commands and tools are used to provide insights into different aspects of the system's memory. Here's a brief overview of memory management and key commands for memory monitoring:

Memory Types in Linux

Physical Memory (RAM):

The actual hardware memory available in the system.

Virtual Memory:

A combination of RAM and disk space (swap), where the system uses disk storage to extend the usable memory.

Swap Space:

Disk space reserved to act as an overflow when RAM is fully used.

Memory Monitoring Commands

free Command:

- The `free` command in Linux is one of the most common tools used for monitoring system memory usage.
- It provides detailed information about the system's total, used, and free memory, as well as the amount of memory being used for buffering and caching.
- It also reports on swap memory usage.

Here's a detailed breakdown of the `free` command:

Syntax of the `free` Command

`free [options]`

Example of `free` Output

When you run `free`, the default output looks like this:

```
$ free
              total        used        free      shared buff/cache   available
```

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Mem:	16349388	10293848	2114220	102400	3947320
	5832048				
Swap:	4194304	2039280	2155024		

Columns Explained

1. **total:** The total amount of memory (RAM) or swap space in the system.
2. **used:** The amount of memory currently in use by the system.
3. **free:** The amount of completely unused memory. This excludes memory used for buffers and caches.
4. **shared:** Memory that is being shared between processes (commonly for shared libraries and IPC).
5. **buff/cache:** The amount of memory used by the system for buffers and caches.
 - **Buffers:** Temporary storage for data as it's being transferred between different hardware components.
 - **Caches:** Memory used to store frequently accessed data for faster retrieval.
6. **available:** An estimate of how much memory is available for starting new applications, without swapping. It takes into account memory used by the kernel for buffers and caches, which can be freed if needed.

Swap Columns:

- **Swap total:** The total amount of swap space (virtual memory) available.
- **Swap used:** The amount of swap space currently being used.
- **Swap free:** The amount of swap space that is still available.

Key free Command Options

1. **-h (Human-readable format)**

The `-h` option displays the memory in a human-readable format (with units like MB, GB):

```
$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	15G	9.8G	2.0G	100M	3.8G	5.5G
Swap:	4.0G	1.9G	2.0G			

The University of Lahore**2. -b, -k, -m, -g (Byte, Kilobyte, Megabyte, Gigabyte display)**

These options specify the unit of memory for the display. By default, `free` displays the output in kilobytes.

- **-b: Show memory in bytes.**

```
free -b
```

- **-k: Show memory in kilobytes (default behavior).**

```
free -k
```

- **-m: Show memory in megabytes.**

```
free -m
```

- **-g: Show memory in gigabytes.**

```
free -g
```

3. -s (Interval option for continuous monitoring)

You can use the `-s` option to specify an interval in seconds for continuous memory monitoring. This will update the memory usage at the specified interval.

```
free -s 5
```

This example updates the memory information every 5 seconds.

4. -t (Display totals)

The `-t` option shows the total memory usage by adding a line with total memory and total swap.

```
free -t
```

Output:

	total	used	free	shared buff/cache	available
Mem:	16349388	10293848	2114220	102400	3947320
	5832048				
Swap:	4194304	2039280	2155024		
Total:	20543692	12333128	4269244		

5. -w (Show detailed information including shared memory and kernel buffers)

The `-w` option provides a wider output including fields related to kernel and shared memory. It gives a deeper look at memory utilization.

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```
free -w
```

6. -c (Count)

The `-c` option lets you specify how many times `free` should display the information before exiting.

```
free -s 2 -c 3
```

Practice Questions

- 1) Display Memory in a Human-Readable Format

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         97Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$
```

- 2) Display Memory Usage in Megabytes

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$ free -m
              total        used        free      shared  buff/cache   available
Mem:          2852         1160          84         40        1808        1692
Swap:         2851           0        2851
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$
```

- 3) Continuous Memory Monitoring with 3-Second Intervals

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$ watch -n 3 free -h
```

```
Every 3.0s: free -h
              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         84Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi
abdullah-sohail-VMware-Virtual-Platform: Thu Mar 26 19:37:50 2026
```

- 4) Monitor Memory Usage Every 2 Seconds and Display Output 5 Times

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$ free -h -s 2 -c 5
              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         80Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi

              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         80Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi

              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         80Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi

              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         80Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi

              total        used        free      shared  buff/cache   available
Mem:          2.8Gi         1.1Gi         80Mi        40Mi        1.8Gi        1.7Gi
Swap:         2.8Gi           0B        2.8Gi
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~$
```

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5) Show Total Memory Usage (RAM + Swap)

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -h -t
```

	total	used	free	shared	buff/cache	available
Mem:	2.8Gi	1.1Gi	82Mi	40Mi	1.8Gi	1.7Gi
Swap:	2.8Gi	0B	2.8Gi			
Total:	5.6Gi	1.1Gi	2.9Gi			

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

6) Display Memory Usage in Bytes

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -b
```

	total	used	free	shared	buff/cache	available
Mem:	2991218688	1214795776	86441984	42434560	1901428736	1776422912
Swap:	2990534656	0	2990534656			

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

7) Show Detailed Memory Information Including Shared Memory and Kernel Buffers

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	2.8Gi	1.1Gi	82Mi	40Mi	1.8Gi	1.7Gi
Swap:	2.8Gi	0B	2.8Gi			

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

8) Memory Monitoring in Gigabytes

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -g
```

	total	used	free	shared	buff/cache	available
Mem:	2	1	0	0	1	1
Swap:	2	0	2			

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

9) Monitor Memory for 3 Intervals and 2 Displays

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -h -s 3 -c 2
```

	total	used	free	shared	buff/cache	available
Mem:	2.8Gi	1.2Gi	114Mi	40Mi	1.7Gi	1.6Gi
Swap:	2.8Gi	0B	2.8Gi			

	total	used	free	shared	buff/cache	available
Mem:	2.8Gi	1.2Gi	114Mi	40Mi	1.7Gi	1.6Gi
Swap:	2.8Gi	0B	2.8Gi			

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

10) Show Memory Usage in Default Kilobytes and Then in Human-Readable Format

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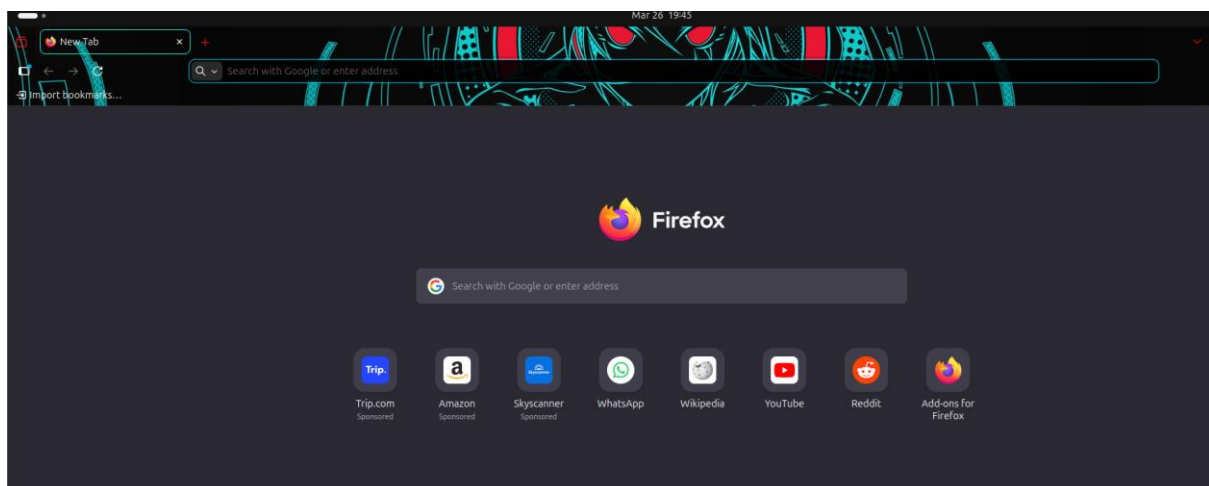
```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free
free -h
              total        used        free      shared  buff/cache   available
Mem:           2921112       1224056        107356        41448       1796268       1697056
Swap:          2920444           0       2920444
              total        used        free      shared  buff/cache   available
Mem:           2.8Gi        1.2Gi        104Mi         40Mi         1.7Gi         1.6Gi
Swap:          2.8Gi           0B         2.8Gi
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

Bonus Challenge Question:

11) Compare Memory Before and After Opening a Large Application

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.8Gi        1.2Gi        104Mi         40Mi         1.7Gi         1.6Gi
Swap:          2.8Gi           0B         2.8Gi
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

Open a heavy app:



```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.8Gi        1.2Gi        486Mi         40Mi         1.3Gi         1.6Gi
Swap:          2.8Gi        316Ki         2.8Gi
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

vmstat Command

- The `vmstat` (virtual memory statistics) command in Linux is used to provide information about system processes, memory, paging, block IO, traps, and CPU activity.

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- It is a valuable tool for performance monitoring and tuning as it gives insights into the current state of the system's resources.

Syntax

vmstat [options] [delay [count]]

- **delay:** Time in seconds between updates.
- **count:** Number of updates.

Key Columns and Their Meaning

When you run `vmstat`, it displays the following fields:

1. procs (Processes):

- **r:** Number of processes in the run queue (waiting for CPU time).
- **b:** Number of processes blocked, waiting for I/O.

2. memory:

- **swpd:** Amount of virtual memory used (in kilobytes).
- **free:** Amount of idle memory (in kilobytes).
- **buff:** Amount of memory used as buffers (for raw disk access).
- **cache:** Amount of memory used as cache (for files).

3. swap:

- **si** (swap in): Amount of memory swapped in from disk (in kilobytes per second).
- **so** (swap out): Amount of memory swapped out to disk (in kilobytes per second).

4. io:

- **bi:** Blocks received from a block device (in kilobytes per second).
- **bo:** Blocks sent to a block device (in kilobytes per second).

5. system:

- **in:** Number of interrupts per second, including the clock.
- **cs:** Number of context switches per second.

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6. **cpu (percentages):**

- **us:** Time spent in user mode (non-kernel processes).
- **sy:** Time spent in system mode (kernel processes).
- **id:** Idle time (percentage of CPU not used).
- **wa:** Time spent waiting for IO operations (idle waiting on disk/network).
- **st:** Time stolen from a virtual machine (hypervisor).

Example Output

```
$ vmstat 2 5
```

```
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-
----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs us sy id
wa st
 1  0       0 31432 12384 145764    0    0   12    4   78  162  2  0 97
1  0
 0  0       0 31424 12384 145764    0    0    0    0   72  148  0  0 100
0  0
 0  0       0 31424 12384 145764    0    0    0    0   64  152  0  0 100
0  0
```

Interpretation:

- **Memory:** This gives you insights into the available and used memory, including the amount being swapped (swpd) or used for buffers and cache.
- **CPU:** The CPU columns show how busy your CPU is with user tasks, system tasks, or if it's spending time waiting for I/O (wa).
- **IO:** High values in bi and bo indicate lots of I/O activity, which could point to disk bottlenecks.
- **Swap:** Large values in si and so indicate swapping is happening, suggesting memory pressure.

Usage Examples:

Basic Usage:

```
vmstat
```

This will provide a snapshot of system performance since the last reboot.

Continuous Monitoring:

```
vmstat 5
```


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This will refresh every 5 seconds and provide a new report.

Limit the Number of Updates:

```
vmstat 2 10
```

This will update the statistics every 2 seconds and stop after 10 iterations.

Practice Questions

- 1) Run `vmstat 2 5` on your system and analyze the CPU usage in the last row of the output. What do the values of `us`, `sy`, `id`, and `wa` indicate?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat 2 5
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r b   swpd   free   buff   cache   si   so    bi    bo    in   cs us sy id wa st gu
 3 0     312 496652   7764 1371612    0    5  2023    77  540    1  2  3 95  0  0  0
 0 0     312 496652   7764 1371628    0    0    0    174  171  0  0 99  0  0  0
 0 0     312 496652   7772 1371620    0    0    12  332  386  0  1  99  0  0  0
 1 0     312 496652   7772 1371628    0    0    0    140  161  0  0 100  0  0  0
 0 0     312 496652   7772 1371628    0    0    4   146  145  0  1  99  0  0  0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

- 2) Your system is experiencing performance issues. Use `vmstat` to check if memory swapping is contributing to the problem. What `vmstat` fields will you focus on, and how will you interpret them?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat 1 5
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r b   swpd   free   buff   cache   si   so    bi    bo    in   cs us sy id wa st gu
 2 0     312 495228   7988 1373152    0    5  1814    71  498    1  2  3 95  0  0  0
 0 0     312 495228   7988 1373160    0    0    0    207  219  0  0 99  0  0  0
 0 0     312 495228   7988 1373160    0    0    0    534  707  2  2  96  0  0  0
 0 0     312 495228   7988 1373160    0    0    0    496  481  1  2  97  0  0  0
 0 0     312 495228   7988 1373160    0    0    0    170  165  1  0  99  0  0  0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

- 3) Use `vmstat` to detect an I/O bottleneck on your system. What fields will you examine, and what patterns indicate a potential I/O issue?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat 1 5
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r b   swpd   free   buff   cache   si   so    bi    bo    in   cs us sy id wa st gu
 3 0     312 436200   8840 1441548    0    4  1469   101  439    1  1  2 96  0  0  0
 0 0     312 435708   8840 1441868    0    0    0    261  230  1  1 99  0  0  0
 0 0     312 435208   8848 1442256    0    0    0    12  260  214  1  0 99  0  0  0
 0 0     312 434716   8848 1442588    0    0    0    238  189  1  1 99  0  0  0
 0 0     312 434716   8848 1442944    0    0    0    841  847  1  2 97  0  0  0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

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- 4) You suspect your system is experiencing high context switching. Use `vmstat` to identify this. What field in the output indicates context switches, and what value would indicate an overhead?

```
vmstat 1 5
system-- ----
rtn      cs us s
27      1 1
74    239 0
77    225 1
46    410 1
85    177 0
```

- 5) Your system shows high values in the `wa` (IO wait) column in `vmstat`. What might this suggest, and what further actions should you take to investigate?

```
vmstat 1 5
system-- -----cpu--
in      cs us sy id wa s
417      1 1 2 96 0
210    226 0 1 99 0
806    861 2 3 96 0
189    165 0 0 99 0
153    157 1 1 99 0
```

- 6) You want to monitor your system's real-time load for the next 30 seconds. Write the command using `vmstat` to achieve this.

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```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat 2 15
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st  gu
 3  0     308 404516   9828 1468712    0    3  1296    98  407    0  1  2  96  0  0  0
 0  0     308 404516   9828 1468712    0    0    0     0  201  205  1  0  99  0  0  0
 1  0     308 404516   9828 1468712    0    0    0     0  375  409  0  1  99  0  0  0
 0  0     308 404516   9828 1468712    0    0    0     0  181  155  0  0  99  0  0  0
 0  0     308 404516   9828 1468768    0    0    0     0  167  176  0  0  99  0  0  0
 0  0     308 404516   9828 1468800    0    0    0     0  431  319  1  1  98  0  0  0
 0  0     308 404292   9836 1468876    0    0    0     6  142  154  0  0  99  0  0  0
 0  0     308 404292   9836 1468932    0    0    0    234  162  161  0  0  99  0  0  0
 0  0     308 404292   9836 1468968    0    0    0     0  211  181  0  0  99  0  0  0
 0  0     308 404292   9844 1469052    0    0    0    10  163  173  0  0 100  0  0  0
 0  0     308 404292   9844 1469084    0    0    0     0  165  166  0  0  99  0  0  0
 0  0     308 404044   9844 1469116    0    0    0     6  425  484  1  1  98  0  0  0
 0  0     308 404044   9852 1469108    0    0    0    12  271  302  0  1  99  0  0  0
 0  0     308 404044   9852 1469228    0    0    0     0  182  170  0  0  99  0  0  0
 0  0     308 404044   9852 1469388    0    0    0     0  192  178  0  0  99  0  0  0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

- 7) What is the difference between **buff** and **cache** in the `vmstat` output? How do these affect system performance?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st  gu
 1  0     308 404044   9892 1469584    0    3  1242    94  398    0  1  2  97  0  0  0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

Disk Monitoring:

- Disk monitoring in Linux involves tracking the performance, usage, and health of disk drives, partitions, and the I/O (input/output) operations they perform.
- Below is a detailed guide covering **basic** to **advanced** disk monitoring commands, including how to interpret the outputs and troubleshoot common issues.

Basic Disk Monitoring Commands

df (Disk Filesystem)

The `df` command displays information about disk space usage on mounted filesystems.

Usage:

df -h

- **-h**: Displays disk usage in human-readable format (MB/GB).

Output Example:

Filesystem Size Used Avail Use% Mounted on

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```
/dev/sda1    50G  20G  28G  42% /  
tmpfs       500M  1.5M  498M   1% /run  
/dev/sdb1   100G  40G  60G  40% /data
```

Key Points:

- **Size:** Total size of the filesystem.
- **Used:** How much space is used.
- **Avail:** How much space is free.
- **Use%:** Percentage of the filesystem used.
- **Mounted on:** The mount point of the filesystem.

du (Disk Usage)

The du command estimates the disk space used by files and directories.

Usage:

```
du -sh /path/to/directory
```

- **-s:** Summarize the total.
- **-h:** Display in human-readable format.

Example:

```
du -sh /var/log
```

This shows the disk usage of the /var/log directory.

lsblk (List Block Devices)

This command lists all available block devices (such as disks and partitions) in a tree format.

Usage:

```
lsblk
```

Example Output:

```
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT  
sda   8:0    0 50G  0 disk  
├─sda1 8:1    0 50G  0 part /  
sdb   8:16   0 100G  0 disk  
└─sdb1 8:17   0 100G  0 part /data
```

This shows the disk devices and their partitions.

The University of Lahore**fdisk (Partition Table Manipulation)**

The fdisk command provides detailed information about disk partitions and allows you to manage them.

Usage:

```
sudo fdisk -l
```

This lists all disks and their partition tables.

Example Output:

Disk /dev/sda: 50 GB, 53687091200 bytes

Device	Boot	Start	End	Blocks	Id	System
--------	------	-------	-----	--------	----	--------

/dev/sda1		2048	97689599	48843776	83	Linux
-----------	--	------	----------	----------	----	-------

mount and umount

- **mount:** View mounted filesystems or mount a device to a directory.
- **umount:** Unmount a mounted filesystem or device.

Usage:

```
mount | grep '/dev/sd'
```

This shows the list of currently mounted disk devices.

iostat (Input/Output Statistics)

The iostat command provides detailed I/O statistics for devices and partitions.

Install:

```
sudo apt-get install sysstat # Ubuntu/Debian
```

```
sudo yum install sysstat # RHEL/CentOS
```

Usage:

```
iostat -x 2 5
```

- **-x:** Display extended statistics.
- **2 5:** Refresh every 2 seconds and display 5 updates.

Example Output:

Device:	rrqm/s	wrqm/s	r/s	w/s	rMB/s	wMB/s	avgrq-sz	avgqu-sz	await	svctm	%util
sda	0.12	0.67	1.03	2.34	0.01	0.02	28.65	0.04	12.33	1.50	5.32

The University of Lahore**Key Metrics:**

- **r/s:** Read requests per second.
- **w/s:** Write requests per second.
- **rMB/s, wMB/s:** Data read and written per second in megabytes.
- **await:** Average time (ms) for I/O requests to complete.
- **%util:** Percentage of time the device was busy (close to 100% indicates saturation).

iotop (Real-Time Disk I/O Monitoring)

iotop is a top-like utility that shows real-time disk I/O usage by processes.

Install:

```
sudo apt-get install iotop
```

Usage:

```
sudo iotop
```

This shows which processes are consuming the most I/O resources.

Key Metrics:

- **DISK READ:** Disk read rate in KB/s.
- **DISK WRITE:** Disk write rate in KB/s.
- **PID:** Process ID using the disk.

smartctl (SMART Monitoring)

smartctl is used to monitor and query the health of storage devices that support SMART (Self-Monitoring, Analysis, and Reporting Technology).

Install:

```
sudo apt-get install smartmontools
```

Usage:

```
sudo smartctl -a /dev/sda
```

Key Output:

- **Temperature:** Shows current device temperature.
- **Reallocated_Sector_Ct:** Indicates the number of reallocated sectors (replacing bad sectors). A high count may signal impending drive failure.

The University of Lahore**dstat (Versatile Resource Statistics)**

dstat is a versatile real-time system resource monitoring tool that combines information from vmstat, iostat, and netstat.

Install:

```
sudo apt-get install dstat
```

Usage:

```
dstat -d
```

This will show disk-related activity like reads/writes in real time.

Example:

```
--dsk/total-
```

```
read writ
```

```
0.12 15.01
```

Practice Questions

- 1) Run the df command to check the disk space usage of your system. What is the total size of your root filesystem, and how much space is available? Use the -h flag to display the result in human-readable format.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            286M  1.8M  284M   1% /run
/dev/sda2        30G   11G   17G  40% /
tmpfs            1.4G    0   1.4G   0% /dev/shm
tmpfs            5.0M   8.0K  5.0M   1% /run/lock
tmpfs            286M  124K  286M   1% /run/user/1000
/dev/sr0          96M   96M    0 100% /media/abdullah-sohail/CDROM
/dev/sr1          6.0G   6.0G    0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

- 2) You want to estimate the size of your /var/log directory. Use the du command to find out the total size of this directory in human-readable format.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo du -sh /var/log
[sudo] password for abdullah-sohail:
324M    /var/log
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

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- 3) Use the lsblk command to list all block devices attached to your system. What is the size of the partition mounted on /data?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0          2:0    1     4K  0 disk
loop0        7:0    0     4K  1 loop /snap/bare/5
loop1        7:1    0   66.8M  1 loop /snap/core24/1499
loop2        7:2    0  245.1M  1 loop /snap/firefox/6565
loop3        7:3    0    74M  1 loop /snap/core22/2292
loop4        7:4    0  251.6M  1 loop /snap/firefox/7836
loop5        7:5    0   73.9M  1 loop /snap/core22/2045
loop6        7:6    0   18.5M  1 loop /snap/firmware-updater/210
loop7        7:7    0   516M  1 loop /snap/gnome-42-2204/202
loop8        7:8    0   16.4M  1 loop /snap/firmware-updater/216
loop9        7:9    0  531.4M  1 loop /snap/gnome-42-2204/247
loop10       7:10   0   91.7M  1 loop /snap/gtk-common-themes/1535
loop11       7:11   0   395M  1 loop /snap/mesa-2404/1165
loop12       7:12   0   15.5M  1 loop /snap/snap-store/1310
loop13       7:13   0   10.8M  1 loop /snap/snap-store/1270
loop14       7:14   0   48.1M  1 loop /snap/snapd/25935
loop15       7:15   0   48.4M  1 loop /snap/snapd/26382
loop16       7:16   0    576K  1 loop /snap/snapd-desktop-integration/343
loop17       7:17   0    576K  1 loop /snap/snapd-desktop-integration/315
sda          8:0    0   30G  0 disk
├─sda1       8:1    0     1M  0 part
└─sda2       8:2    0   30G  0 part /
sr0         11:0    1   95.3M  0 rom  /media/abdullah-sohail/CDROM
sr1         11:1    1    5.9G  0 rom  /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

- 4) Run the fdisk command to display partition information for all disks. Which partition has the boot flag enabled, and what is its size?


```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo fdisk -l  
Disk /dev/loop0: 4 KiB, 4096 bytes, 8 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
  
Disk /dev/loop1: 66.83 MiB, 70078464 bytes, 136872 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
  
Disk /dev/loop2: 245.13 MiB, 257036288 bytes, 502024 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
  
Disk /dev/loop3: 73.96 MiB, 77549568 bytes, 151464 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
  
Disk /dev/loop4: 251.64 MiB, 263868416 bytes, 515368 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

- 5) Use the mount command to list all currently mounted filesystems. Which device is mounted on /data?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mount | grep /data
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

- 6) Use the `iostat` command to check the I/O statistics of your disk. How many read requests per second (r/s) and write requests per second (w/s) are happening on `/dev/sda`? Run the command with extended statistics and display the results every 2 seconds for 5 iterations.

[illegible]

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- 7) Use the iotop command to monitor real-time disk I/O usage by processes. Identify the process that is consuming the most disk read bandwidth.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo iotop
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
Total DISK READ:      0.00 B/s | Total DISK WRITE:      0.00 B/s
Current DISK READ:    0.00 B/s | Current DISK WRITE:    0.00 B/s
  TID  PRIO  USER      DISK READ  DISK WRITE  COMMAND
  ----  ----  -
    1  be/4  root       0.00 B/s   0.00 B/s   systemd --deserialize=123 splash
    2  be/4  root       0.00 B/s   0.00 B/s   [kthreadd]
    3  be/4  root       0.00 B/s   0.00 B/s   [pool_workqueue_release]
    4  be/0  root       0.00 B/s   0.00 B/s   [kworker/R-rcu_gp]
    5  be/0  root       0.00 B/s   0.00 B/s   [kworker/R-sync_wq]
    6  be/0  root       0.00 B/s   0.00 B/s   [kworker/R-kvfree_rcu_reclaim]
    7  be/0  root       0.00 B/s   0.00 B/s   [kworker/R-slub_flushwq]
    8  be/0  root       0.00 B/s   0.00 B/s   [kworker/R-netns]
   12  be/4  root       0.00 B/s   0.00 B/s   [kworker/u512:0-ipv6_addrconf]
   13  be/0  root       0.00 B/s   0.00 B/s   [kworker/R-mm_percpu_wq]
   14  be/4  root       0.00 B/s   0.00 B/s   [ksoftirqd/0]
   15  be/4  root       0.00 B/s   0.00 B/s   [rcu_preempt]
   16  be/4  root       0.00 B/s   0.00 B/s   [rcu_exp_par_gp_kthread_worker/1]
   17  be/4  root       0.00 B/s   0.00 B/s   [rcu_exp_gp_kthread_worker]
   18  rt/4  root       0.00 B/s   0.00 B/s   [migration/0]
   19  rt/4  root       0.00 B/s   0.00 B/s   [idle_inject/0]
   20  be/4  root       0.00 B/s   0.00 B/s   [cpuhp/0]
   21  be/4  root       0.00 B/s   0.00 B/s   [cpuhp/1]
   22  rt/4  root       0.00 B/s   0.00 B/s   [idle_inject/1]
   23  rt/4  root       0.00 B/s   0.00 B/s   [migration/1]
   24  be/4  root       0.00 B/s   0.00 B/s   [ksoftirqd/1]
   27  be/4  root       0.00 B/s   0.00 B/s   [cpuhp/2]
   28  rt/4  root       0.00 B/s   0.00 B/s   [idle_inject/2]
   29  rt/4  root       0.00 B/s   0.00 B/s   [migration/2]
   30  be/4  root       0.00 B/s   0.00 B/s   [ksoftirqd/2]

keys: any: refresh  q: quit  i: ionice  o: active  p: procs  a: accum
sort:  r: asc  left: DISK READ  right: COMMAND  home: TID  end: COMMAND
CONFIG_TASK_DELAY_ACCT and kernel.task_delayacct sysctl not enabled in kernel, cannot determine SWAPIN and IO %

```

- 8) Run the smartctl command to check the health status of /dev/sda. What is the current temperature of the disk, and are there any reallocated sectors?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo smartctl -a /dev/sda
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.17.0-14-generic] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:                 VMware,
Product:                VMware Virtual S
Revision:               1.0
User Capacity:          32,212,254,720 bytes [32.2 GB]
Logical block size:     512 bytes
Rotation Rate:          Solid State Device
Device type:             disk
Local Time is:          Thu Mar 26 20:29:21 2026 PKT
SMART support is:        Unavailable - device lacks SMART capability.

=== START OF READ SMART DATA SECTION ===
Current Drive Temperature:     0 C
Drive Trip Temperature:        0 C

Error Counter logging not supported

Device does not support Self Test logging
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

- 9) Use the `dstat` command to monitor disk reads and writes in real time. Capture the read and write activity on your system for a few seconds.

```

Mar 26 20:34
(abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform) dstat -d
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ dstat -d
dsk/total
  read  writ
    0     0
    0     0
4106B  160k
    0  4091k
    0     0
    0     0
    0     0
    0  108k
    0     0
    0     0
    0     0
    0  768k
    0  4096B
    0   40k
    0     0
    0   96k
    0     0
    0     0
    0     0
    0   60k
    0   36k
    0  332k
    0     0

```

- 10) Monitor the disk I/O statistics using `iostat -x`. After analyzing the output, determine the average time (await) for an I/O request to complete on `/dev/sda` and the disk utilization percentage (%util).

[illegible]